Design and Evaluation of an AI-Based American Sign Language Video Comprehension Tool: Exploring the Benefits and Use of Automatic Segmentation and Sign Lookup

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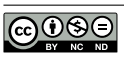
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Abstract

Sign language learners often watch continuous signing videos to develop their receptive and expressive skills, yet looking up and identifying unfamiliar signs during these comprehension tasks remains a challenge. Prior HCI research has explored in-situ sign lookup tools using Wizard-of-Oz prototypes, enabling learners to search for signs by submitting video segments as queries. Findings showed that this approach improved comprehension and reduced cognitive load compared to feature-based dictionaries. Building on advances in AI-based sign language recognition and segmentation, we present an automatic video comprehension prototype system that segments continuous signing into individual signs and employs a fine-tuned recognition model to retrieve the closest matches for each segment. We report findings from an experimental study with thirteen intermediate ASL learners, combining quantitative and observational analyses to compare our system with a baseline that supports user-selected segment-based search but lacks AI-based automatic segmentation. Results show that automatic segmentation improved user experience, while observational data revealed distinct interaction patterns between the two prototype systems. We release both prototype systems and present design guidelines for video comprehension tools for sign language learners.

CCS Concepts

• **Human-centered computing** → *Empirical studies in accessibility*; *Accessibility technologies*; **Empirical studies in HCI**; **Interactive systems and tools**.



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Keywords

Sign Language, American Sign Language, Language Learning, Sign Language Learning, Artificial Intelligence, Sign Language Recognition, Sign Language Segmentation, Human-AI Interaction, Video Comprehension

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1 INTRODUCTION

American Sign Language (ASL) is the primary language for about 500,000 people in the U.S. [46] and is also used by many in Canada [19]. Beyond the Deaf and hard of hearing (DHH) community, many hearing people, including parents, caregivers, interpreters, and researchers, are also motivated to learn ASL [54]. Enrollment in ASL and Deaf Studies courses continues to grow rapidly, and ASL has become the third most studied language in the U.S. [40, 43], with many higher education institutions now offering multiple sections of introductory courses to meet this demand [1, 12]. To support this growing demographic, this paper focuses on the needs of hearing intermediate learners who may navigate aspects of ASL learning independently, without continuous access to native signing environments.

ASL is a linguistically complex language and is often considered *one of the most difficult languages to learn* [33], in part because it differs fundamentally from spoken and written languages. Its grammar and syntax are distinct from English, and use spatial structures, non-manual markers (e.g., facial expressions and body movements), and classifier constructions[1] to convey meaning [45]. ASL also relies on intricate morphological processes such as compound signs,

[1]Classifier constructions, or classifier predicates, are a morphological system for expressing events and states in sign languages [45]. Handshapes, movement, and spatial location could be used to represent nouns and verbs, and enable concise and iconic descriptions of actions or states. For example, a "V" handshape can depict two people walking, with its movement indicating how they walk.

where distinct signs combine into a single, fluid motion. Learners must also develop strong visual-spatial processing skills, including the ability to perceive and reproduce signs in three-dimensional space [69]. With limited opportunities to practice with fluent signers, *many learners turn to sign language videos as a learning resource.*

A major challenge in video-based ASL learning is the limited availability and poor usability of tools that effectively support this mode of independent learning. Most existing technological resources for ASL learning focus primarily on basic vocabulary and rarely extend beyond helping learners recognize isolated signs in their "citation" form[2]. These tools are typically either vocabulary learning games (e.g., [6, 34, 61]) or feature-based dictionaries (e.g., [10, 38]) that allow learners to look-up an unfamiliar sign using linguistic properties. However, classifying signs by linguistic properties can be especially challenging for hearing beginner and intermediate sign language learners. For example, to search for the sign Book using a commercially available feature-based dictionary [38], a user must know that its handshape is characterized by "5-close, 15," involves a *repeated movement*, occurs at the *palm location*, and uses a *two-handed symmetrical configuration*. Even if learners are familiar with or attempt to infer these linguistic properties from context, *they must leave the video viewing environment to perform the search, which can be disruptive*. By contrast, second-language learners of spoken or written languages can simply highlight an unfamiliar word in text or use captions and transcripts to look up its meaning seamlessly. Currently, no commercially available tools provide comparable lookup or comprehension support for ASL videos.

A recent study used a Wizard-of-Oz (WoZ) approach to investigate a tool that allows learners to look up the meaning of a sign within a video segment selected using a playhead [23]. Findings from this work show that ASL learners not only improve their comprehension but also experience reduced workload when looking up complex signs compared to using traditional feature-based approaches. Recent advances in sign segmentation [47, 56, 62] and recognition [14, 75] now make it possible to automate this tool. Of particular relevance is the emergence of automatic sign segmentation [47], which has the potential to provide learners with video segments and the closest matching signs for each segment. However, it remains unclear whether AI-generated automatic segmentation is genuinely helpful for ASL learners and to what extent they prefer it over manually segmenting videos to submit search queries. More importantly, designing a fully automatic tool allows HCI researchers to examine how learners interact with imperfect AI outputs that are difficult to simulate, such as inconsistent performance or incorrect sign retrieval. Studying these scenarios can reveal how users adapt their behavior, modify their searches, and how automatic segmentation influences their translation process and engagement with sign language videos.

[2]In sign linguistics, a sign in "citation form" refers to its standard or dictionary form, produced in isolation rather than within a sentence or natural discourse. This version shows the sign's canonical handshape, movement, location, and orientation, without the modifications that occur in context (such as changes for grammatical agreement, emphasis, or coarticulation) [63].

- We present and release an automatic ASL video comprehension prototype system incorporating sign language segmentation, building on the user interface from prior Wizard-of-Oz work [23]. Its underlying AI pipeline pairs a state-of-the-art sign segmentation model with a fine-tuned recognition model. Although limited in vocabulary and not a full continuous segmentation and recognition solution for the entirety of ASL, it provides a practical foundation for interactive video-based lookup.
- We report findings from an experimental study with thirteen ASL learners, comparing their experiences using our **Segment and Search (S&S)** prototype system, which integrates segmentation and search functionality as well as recognition of user-selected video inputs, with a **Search Only (S)** baseline prototype system that includes all other features but supports only recognition using user-selected inputs. Our findings show that participants reported lower workload and higher search satisfaction with the S&S prototype, while translation-quality scores were numerically higher but did not differ significantly between conditions. Observational analyses further revealed distinct interaction patterns across the two conditions, particularly in the duration and manner of using the span selector.

The contributions of this work are both empirical and artifact. We release the prototype systems alongside this publication (shared as supplementary material) and present design guidelines for future video-based ASL comprehension tools, as well as implications for broader research on video understanding technologies.

2 Background and Related Work

In this section, we discuss prior work in two areas: HCI research on sign language lookup tools and AI research on underlying recognition and segmentation methods. We situate our contributions within the existing literature and highlight how our work builds upon and extends these efforts.

2.1 Human-Computer Interaction Research on ASL Look-up Tools for Learners

Extensive HCI research has explored sign language learning technologies, with much of this work focusing on tools that support the learning of isolated signs. Prior studies have examined the design of digital dictionaries [7, 10, 24, 36] and game-based learning systems [6, 61] that help learners improve recognition and recall of individual signs through interactive feedback and repetition.

Dictionaries or **word look-up** tools are essential for learning any language, including sign languages. Sign language dictionaries can exist in print (e.g., [64]) or digital formats (e.g., [7, 10, 38, 59]). Bragg et al. [10] conducted an open-ended survey with ASL learners about their use of ASL dictionaries, which revealed the need for a user-friendly and reliable web-based tool for ASL sign lookup. The study also highlighted that existing resources for finding English translations of ASL signs were rarely used. Current ASL dictionaries proposed in research employ a variety of input modalities, which can be broadly grouped into two categories: *search-by-feature* and *search-by-video*. Several feature-based ASL dictionaries are commercially available [38, 59] or have been developed by researchers

[10, 36]. These systems allow users to filter signs based on one or more linguistic features. Unlike video-based dictionaries, they do not require users to record or upload videos, thus avoiding privacy concerns and latency issues. However, as mentioned in the introduction, they rely on users to decompose a sign they encountered into its linguistic components, often represented through formal notations, e.g., Stokoe notation [63]. This process can be particularly difficult for novice learners, who may struggle to recall or analyze a sign's form. Unlike written languages, which allow for spelling approximations, ASL's visual-spatial nature requires users to input precise parameters to locate a sign. Consequently, research has shown that ASL learners rarely use feature-based dictionaries to look up unfamiliar signs [11, 24].

Recent advances in sign recognition technology have enabled the development of *search-by-video* ASL dictionaries. In particular, these video-based approaches often leverage advanced video editing and manipulation techniques to process continuous signing into searchable segments [55]. These tools allow users to submit video samples of unknown signs without specifying linguistic features, making them especially useful when users have a video of the sign or can reproduce it from memory. Although model performance has improved over time, sign recognition remains a challenging task due to the difficulty of interpreting three-dimensional signs from two-dimensional video input [9, 70]. Persistent issues include limited recognition accuracy and challenges in human–AI interaction (HAI), particularly when users struggle to capture high-quality video recordings and struggle with understanding the quality of the AI-generated output.

To address the challenge of requiring users to record new videos, prior HCI work has proposed allowing learners to use existing video samples, e.g., clips from longer ASL videos they already possess, as input for sign lookup. This approach can be particularly beneficial in scenarios such as video comprehension or translation tasks, where learners frequently encounter unfamiliar signs within recorded material. In work closely related to ours, Hassan et al. [23] explored a system that enables ASL learners to input segments of ASL videos to look up unknown signs encountered during video watching or transcription. Their findings showed that this method improved transcription quality and reduced user workload compared to a *search-by-feature* dictionary [38]. The study also revealed distinct interaction patterns, e.g., users employed the video playhead not only to define lookup segments but also to navigate and review signing progressively. Additionally, behavioral differences emerged across video genres, e.g., conversational versus theatrical signing.

Our work builds on this prior research in two key ways. **First**, the earlier system relied on a WoZ approach, in which the recognition outputs were simulated using an estimated protocol to emulate imperfect AI performance. Sign segmentation, on the other hand, was based on manual annotations by a Deaf researcher, and users were not provided with an option to segment selected spans. With recent advances in both sign segmentation [30, 47] and recognition [14, 49, 75], it is now possible to evaluate such systems using real AI models. Evaluating an automatic tool can not only help assess the performance of state-of-the-art underlying technologies but also reveal usage patterns and new human–AI interaction challenges, e.g., how users engage with automatic recognition both with and without segmentation to understand video content. It can also help evaluate how participants respond when the output becomes less consistent and how they adjust their use of the span to compensate for imperfect system performance. **Second**, the prior study compared the *search-by-video-span* lookup system only against a *search-by-feature* baseline (the Handspeak reverse dictionary [38]). While this helped demonstrate the advantages of *search-by-feature* lookup over similar approaches, it does not evaluate the benefits of automatic sign segmentation compared to *search-by-video*. In our study, we use a *search-by-video-span*, or Search-only (S), baseline in which users can manually select segments from videos for lookup. This allows us to better understand how ASL learners engage with automated sign segmentation and how it influences their comprehension experience, informing both computer vision researchers working on automatic segmentation and designers of these comprehension systems.

2.2 AI Research on Sign Language Recognition and Segmentation

Designing video-based sign language dictionaries requires automatic sign language recognition [9]. Over the years, there has been significant progress in sign language recognition with the majority of work focusing on ASL. This work has been driven by advances in both *model architectures* and the development of *large-scale datasets* for training.

Historically, research has focused primarily on isolated sign language recognition [9]; however, recent efforts have expanded toward Continuous Sign Language Recognition (CSLR), which involves identifying successive signs in a video stream without prior knowledge of their temporal boundaries [4]. Despite these advances, CSLR remains an open challenge due to limited data availability, restricted sign language vocabularies, and complex linguistic phenomena, e.g., coarticulation [58] and classifiers [3]. Even if CSLR eventually achieves the accuracy[3] needed to automatically recognize continuous signing videos, *learners developing their skills would still benefit from being able to look up individual signs and analyze shorter segments of signing*. To enable such learning-focused applications, a closely related technology, *sign language segmentation*, is required. It involves breaking down continuous signing videos into individual glosses that represent distinct signs or grammatical constructions. In this section, we summarize the state of the art in both *automatic isolated sign recognition* and *sign language segmentation*.

Early research on sign language recognition relied heavily on handcrafted features and small, controlled datasets, e.g., [73] and [44], where models often used Support Vector Machines (SVMs) or Hidden Markov Models (HMMs) for classification [60]. The introduction of depth sensors, e.g., Microsoft Kinect around 2010, enabled multimodal datasets that incorporated RGB, depth, and skeleton data, marking an important shift toward richer visual representations. With the rise of deep learning, 2D-CNNs and RNNs dominated early work between 2015 and 2018, before being gradually replaced by 3D-CNNs and Transformer-based architectures that captured spatiotemporal dependencies in video data more effectively [48, 68].

[3]Several foundation models for full sign language recognition, e.g., SignGemma [18] and SignX [16], have recently been introduced or are expected to be released soon.

At the same time, datasets evolved from small, laboratory-controlled collections to large-scale, diverse corpora, e.g., *MS-ASL* [32], *WLASL* [67], and *ASL-RGBD-100* [25], which enabled more generalized and transferable models across signers and contexts. These developments set the stage for pose-based and Transformer-based methods, e.g., the Sign Pose Transformer (SPOTER), which achieved state-of-the-art performance while maintaining computational efficiency suitable for mobile and real-time applications [8].

In recent years, sign language recognition has advanced with large (both in terms of samples and number of signs) and diverse open-source datasets, e.g., *ASL Citizen* [14]. Recent Transformer-based recognition models trained on this dataset have achieved state-of-the-art performance, benefiting from large-scale pretraining and improved temporal modeling strategies [14]. However, these models have primarily been trained and evaluated on isolated signs performed in their "citation form" or lab-based annotated and pre-segmented video clips rather than on outputs from automatic sign segmentation pipelines. Early efforts have begun to train and evaluate sign-recognition models on the outputs of automatic or linguistically annotated segmentation pipelines. For example, Zhou et al. [75] demonstrated that combining skeleton- and handshape-based Graph Convolutional Networks (GCNs) with temporal boundary detection modules improves accuracy for both isolated and pre-segmented signs.

Recent advancements in *sign language segmentation* have focused on improving temporal boundary detection between consecutive signs and phrases using both visual and linguistic cues. Earlier approaches, e.g., by Renz et al. [56] combined pseudo-labelling and changepoint detection to iteratively refine temporal segments in continuous signing videos, improving generalization without requiring fully annotated data. Building on this, Moryossef et al. [47] proposed a *linguistically motivated segmentation* framework that models segmentation at both the sign and phrase levels using a *BIO-tagging scheme*, optical-flow-based prosodic features, and 3D hand normalization. This approach explicitly integrates linguistic information, e.g., prosody and handshape transitions, to identify phrase and sign boundaries, outperforming earlier pose-based and statistical methods while demonstrating strong cross-linguistic generalization. Most recently, Stassi et al. [62] presented a comparative analysis of state-of-the-art segmentation methods, confirming that linguistically motivated segmentation achieves the most consistent and interpretable results across datasets in different sign languages.

In **our work**, we employ state-of-the-art approaches for isolated sign recognition [14, 75] and leverage large datasets [14, 50, 51] to fine-tune a recognition model capable of handling signs in both their automatically segmented and citation forms. For sign segmentation, we adopt the latest linguistically motivated segmentation approach [47] and evaluate its performance on our stimuli videos used in our experimental study.

2.3 Focus and Research Question

We present and release two AI-based video comprehension tools for ASL learners: one with automatic segmentation (S&S) and one without (S). These tools enable learners to look up unfamiliar signs within sign language videos. Our experimental study compares how intermediate learners use these prototypes while translating ASL into English sentences:

RQ When comparing the experiences of intermediate ASL learners using an *in-situ* video-comprehension tool **with** and **without** AI-based automatic segmentation while closely watching and translating ASL videos, are there differences in their:

1. perceived workload,
2. reported satisfaction,
3. translation quality, and
4. use of user-interface (UI) elements, e.g., span selector, in the prototype systems?

3 System Design

In this section, we describe the design of our two prototype systems (S&S and S) detailing both the user interface and the underlying AI pipeline, which consists of automatic sign segmentation and recognition.

3.1 User Interface

The UI of both prototype systems builds upon the design described in [23] and is inspired by HCI research on ASL dictionaries [10, 24, 26]. As shown in Figure 1, both web-based systems consist of five main components: a VIDEO PLAYER SCREEN, TRANSCRIPTION PANEL, SEARCH RESULTS PANEL, VIDEO TIMELINE, and SPAN SELECTION functionality. To support additional interactions, we added four user-interface elements: a SEGMENT AND SEARCH button, a SEARCH button, and UPLOAD VIDEO and RECORDING buttons[4].

In the **Segmentation and Search** (S&S) prototype, the SEGMENT AND SEARCH button is added to allow users to trigger automatic segmentation and recognition on segmentation output. After selecting a span on the video timeline using the adjustable span selector PLAY HEAD by dragging the start and end markers, users can trigger the segmentation system to divide the selection into individual signs. In practice, this selection can be the entire duration of the video or a small segment that the user wants to analyze.

Once the user clicks SEGMENT AND SEARCH and processing is complete, a confirmation message appears: PROCESSING DONE, SHOWING 20 RESULTS BETWEEN SELECTED START TIME–END TIME. Each automatically detected sub-segment is displayed in the navigation bar (Figure 2) in chronological order, and selecting a sub-segment highlights the corresponding region on the timeline. This allows users to quickly assess recognition quality and compare retrieved signs to their target signs without leaving the video context. The retrieved results are shown in the SEARCH RESULTS panel (Figure 1), where each sign is represented as a looping GIF along with a GLOSS label[5] and confidence score. Confidence levels are generated based on the system confidence or probability of a sign being the correct

[4]These two buttons were not used in the study but are included in the released prototype to enable future functionality, such as uploading or recording videos.

[5]An ASL gloss is a system of written English words in capital letters that label ASL signs, serving as an interlinear translation tool to represent ASL's visual-spatial grammar and structure. It is not a true written form of ASL, which is a visual language, but a way for learners and linguists to analyze and discuss signs [66].

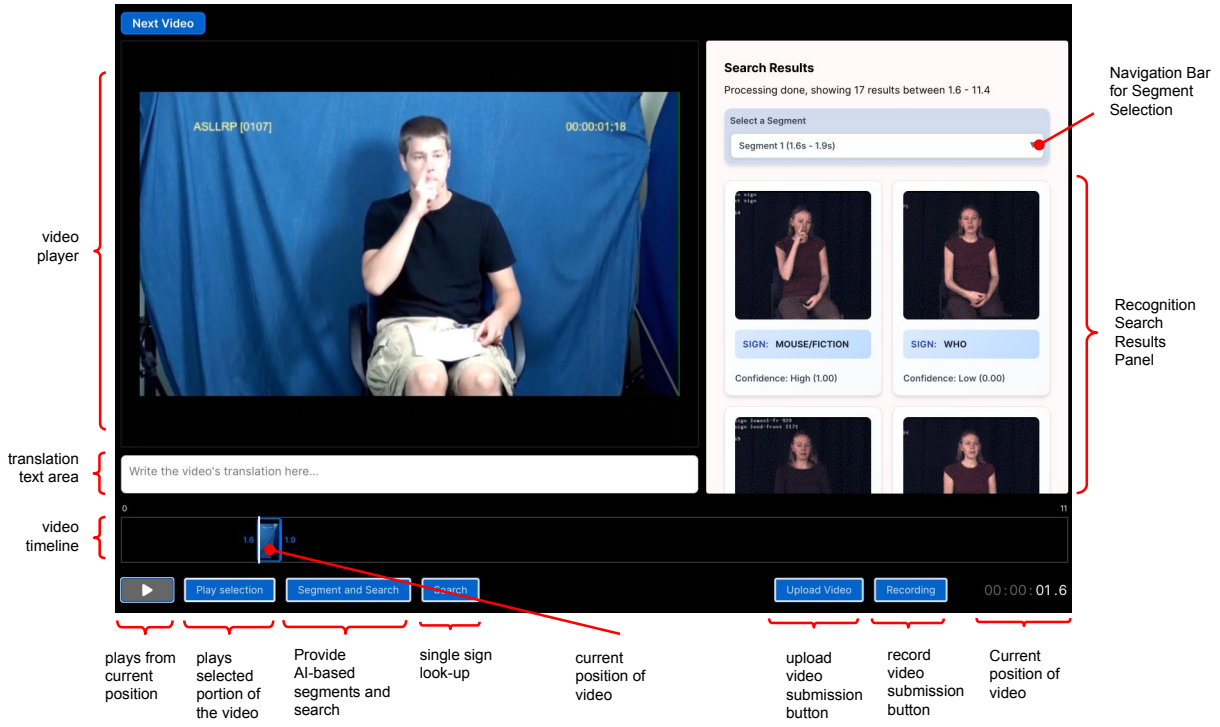


Figure 1: An overview of the prototype interface designed to support ASL video comprehension. The layout includes the main video player (top-left), a span-selection and control timeline for segmenting the video (bottom), and the results panel (right) showing the output of automatic segmentation and recognition.

sign and are grouped into three ranges: Low (0–33%), Medium (34–66%), and High (67–100%). This presentation of individual result snippet is inspired by prior HCI research [7, 26].

In contrast, the **Search-Only prototype** (S) includes only the SEARCH button. This version omits automatic segmentation, requiring users to manually select a subsegment corresponding to a single sign on the timeline before retrieving the top-20 AI-based recognition results for the selection. This system lacks the ability to automatically suggest sign boundaries or present multiple recognition results for segmented regions within a selected video span. In practice, if a user selects a subsegment containing multiple signs, the recognition performance, as indicated in the results, is expected to decrease substantially, which may encourage the user to select a finer subsegment using the play head.

3.2 Segmentation Model

For automatic segmentation, we used the linguistically motivated segmentation model proposed in [47]. This model adopts a BIO-tagging scheme (Beginning–Inside–Outside) to identify the start and end points of each sign in continuous signing. The approach is grounded in linguistic cues such as prosody, motion, and handshape changes, and jointly models both sign- and phrase-level boundaries using pose-based features extracted from the signer's body and hands. Because it relies on motion and structural patterns rather than language-specific gloss annotations, the model is expected to generalize well across different sign languages[6].

We evaluated the model's performance on a broader corpus of 45 continuous ASL videos (ranging in length from 1 to 11 minutes) taken from [50]. These videos consist of a single signer signing in front of a uni-color background.

In consultation with the ASL educator on our team, we further narrowed this set to four final stimuli videos used in the user study [50]. Our ASL educator noted that intermediate learners typically require approximately 15–20 minutes to translate a 10-second video—an estimate that our study subsequently confirmed. Therefore, we selected four 10-second stimulus videos featuring three different signers across both conversational and narrative signing styles, encompassing 39 unique signs (278 gloss labels total).

Video 1 depicted a narrative of mouse escaping from a cat, with signs such as MOUSE/FICTION, ESCAPE, CAT, CHASE, TABLE, and MONEY. **Video 2** consisted of an educational scenario involving teaching and culture, with signs like TEACH+AGENT, STUDENT, BOOK, NAME, SHOW, and CULTURE. **Video 3** was based on a daily routine with a roommate and morning activities, featuring signs like ROOMMATE, PAST+NIGHT, BREAKFAST, WEATHER, and LOUSY.

[6]We also experimented with retraining the model on annotated, language-specific segmentation data, but found that its performance degraded. In contrast, the original pretrained model exhibited graceful degradation, suggesting stronger generalization across sign languages.

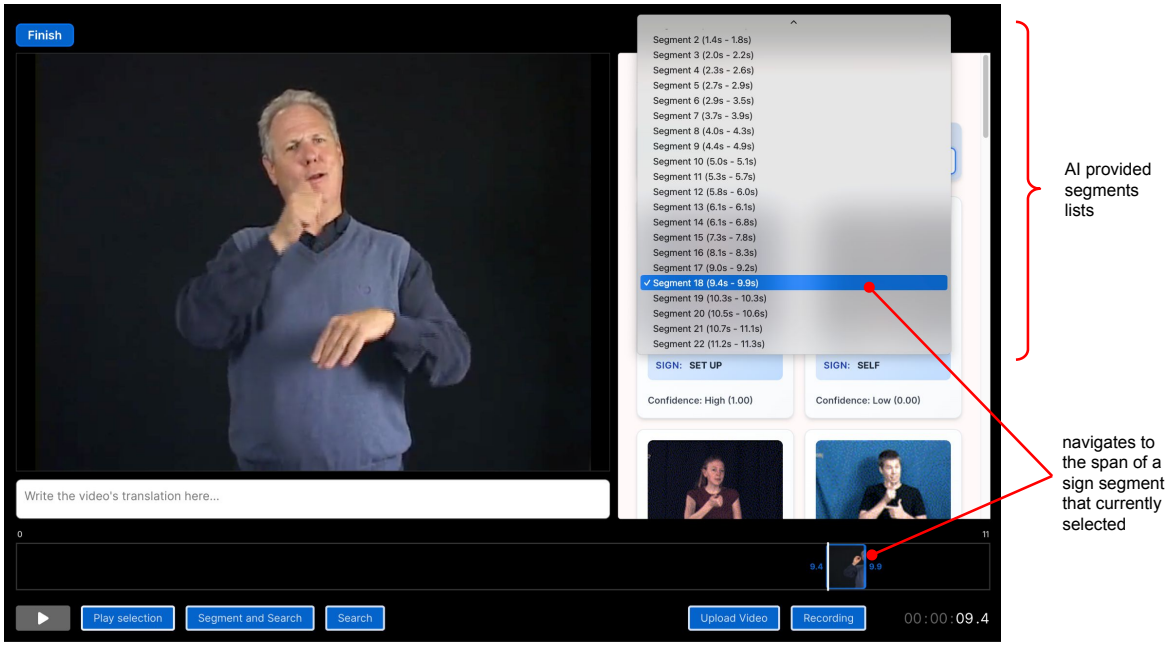


Figure 2: The segment navigation bar displays the output of the automatic segmentation model. Each segment block represents an individual sign detected by the AI within the user's selected span. Users can click a segment to highlight its corresponding region on the video timeline and view the specific recognition results in the Search Results Panel.

Video 4 presented a conditional situation about applying for a job, including signs like IF, SEND, RESUME, GET, and JOB.

We used two metrics introduced in the original work [47] to report findings from our evaluation of segmentation model:

- **Intersection over Union** (IoU): For each video, we calculate the IoU for all matched segment pairs of automatic segmentation and human-annotated labels in the datasets. Each predicted segment $p_i = [s_p, e_p]$ is compared to its corresponding human-annotated ground-truth segment $g_j = [s_g, e_g]$. Then, we compute the Intersection over Union (IoU) as the ratio between the overlapping duration (intersection) and the union duration of the two intervals:

$$\text{IoU}(p_i, g_j) = \frac{\text{Length of Overlap}(p_i, g_j)}{\text{Length of Union}(p_i, g_j)} = \frac{|p_i \cap g_j|}{|p_i \cup g_j|} \quad (1)$$

- **Over-Segmentation Ratio** (Over-segmentation %): We compute this metric to measure how many segments the model predicts compared to the number of human-annotated or ground-truth segments. The optimal value is 1, indicating that the model predicts the same number of segments as the ground truth.

$$\frac{\#\text{ of Predicted segments}}{\#\text{ of Ground Truth segments}} \quad (2)$$

Table 1 summarizes our findings. According to Moryossef et al. [47], the original model achieved an IoU of 0.69 and an over-segmentation percentage of 1.03 on [52]. On our dataset of 45 continuous signing videos, the model achieved an IoU of 0.37 and an over-segmentation percentage of 1.09. On the 10-second prototype stimuli videos, the model achieved an IoU of 0.50 and over-segmentation percentage of 1.21.

Table 1: Segmentation performance across datasets and segmentation conditions (mean ± standard deviation).

Dataset	Over-segmentation %	IoU
Moryossef et al. [47]	1.03	0.69
ASLLRP [50]	1.09 ± 0.71	0.37 ± 0.23
Stimuli Video	1.21 ± 0.15	0.50 ± 0.19

3.3 Recognition Model

3.3.1 Training Data. We selected 991 glosses from the American Sign Language Lexicon Video Dataset (ASLLVD) [51] and the American Sign Language Linguistic Research Project (ASLLRP) [50] datasets to train a recognition model compatible with the outputs of our segmentation model in addition to signs in their "citation form." The ASLLVD dataset provides isolated sign segments extracted from continuous signing videos, whereas the ASLLRP dataset consists of signs produced in citation form. We first selected 549 glosses from these datasets with at least nine example videos each, and then added 442 glosses to expand coverage.

3.3.2 Recognition Pipeline. For our isolated sign recognition model, we used the Spatial-Temporal Graph Convolutional Network (ST-GCN) architecture [14]. ST-GCN is a pose-based model that offers

competitive performance compared to heavier architectures like the Inflated 3D Convolutional Network (I3D). Crucially, it provides significantly faster inference speed. In our system, ST-GCN processed a 10-second video in under 30 seconds, compared to over 1 minute for I3D, making it well-suited for integration into our real-time prototype system.

Starting with weights pretrained on the ASL Citizen dataset [14], we fine-tuned the model on our dataset of 991 glosses for 200 epochs using Stochastic Gradient Descent (SGD) [57] with a learning rate of 1×10^{-1}. Our dataset exhibited substantial class imbalance, with gloss instances ranging from 1 to 289 per class. To address this, we used a weighted random sampler that assigned higher sampling probabilities to underrepresented glosses.

Following the overall model pipeline described in [14], we extracted 27 keypoints from the hands and face using MediaPipe Holistic [42], then center-scaled and normalized them. We applied data augmentation techniques, including random rotation and shear transformations on pose sequences (described in the original paper [14]). For optimization, we used cross-entropy loss with a cosine annealing learning rate scheduler [41] ($T_{max} = 150$). The encoder was initialized with pretrained weights, and the classifier was reinitialized using Xavier initialization [13].

3.3.3 Performance Evaluation. To assess the utility of our automatic segmentation prototype, we evaluated how well the sign recognition model handles the output of our automated segmentation. The goal of this analysis was to identify segments of videos where both AI models performed reasonably well, as our primary objective was to evaluate the usability and human-AI collaboration aspects of automatic segmentation rather than to benchmark or optimize the underlying models themselves.

According to Desai et al. [14], ST-GCN achieved 60% top-1 and 88% top-10 accuracy on the ASL Citizen benchmark. Our retrained model on 991 glosses reached 69% top-1 accuracy on the validation set, which provides a reference for in-domain performance under clean segmentation. Ultimately, all data, including the user study stimuli, was included in the final training to maximize the usability of user study. We report the performance of our trained recognition model at two levels: (1) generalization to a broader corpus of 45 continuous signing videos (ranging from 1 to 11 minutes in length) and (2) performance on the 10-second prototype stimuli videos used in our user study (described in Section 3.2).

For the broader corpus, we compared recognition performance when using automatically generated segments from our segmentation model output versus ground-truth (GT) segments (released along with the dataset our stimuli videos were taken from [49]). As shown in Table 2, on average, AI-segmented spans achieved 33% (top-1), 39% (top-2), and 51% (top-20) accuracy, whereas GT-segmented spans performed better with 46%, 50%, and 56%, respectively[7]. Finally, we evaluated the model on 10-second prototype stimuli videos that were used in our user study. These clips achieved 61% (top-1), 68% (top-2), and 91% (top-20) accuracy on AI-segmentation outputs, which shows that the recognition performance degrades gracefully as compared to human-annotations.

[7]Median comparisons of top-1, top-2, and top-20 results: AI-segmentation yielded 50%, 61%, and 83% at top-1, top-2, and top-20, while GT-segmentation achieved substantially higher performance of 73%, 80%, and 92%.

This performance discrepancy between the broader corpus and the stimuli videos is likely driven by the varying linguistic complexity and sign composition across the datasets. The broader corpus contains several signs with very short durations (e.g., NOW), and grammatical or sentence structures, like negations (e.g., NOTHING) that current recognition models frequently miss in segments from continuous videos. In contrast, the selected 4 videos contained a higher proportion of salient, well-defined motion trajectories associated with common sentence structures and clearly articulated signs (e.g., TEACHER) which led to higher performance than the broader corpus. We anticipate that in actual classrooms or real-world applications, performance may degrade further.

4 Experimental Study Methods

After designing the two prototype systems, we conducted a lab-based experimental study to compare them on measures of performance and usability. Our goal was to understand whether automatic segmentation was helpful and how it affected participants' video-viewing and translation-writing behaviors.

4.1 Participants

We recruited participants by posting flyers on university bulletin boards and contacting professors who teach advanced ASL courses.

The participant group was selected in consultation with the ASL educator on our team. Our goal was to recruit participants with **intermediate ASL proficiency**. For novice learners, translating long videos would have been too challenging even with the tool's support, whereas for expert signers, the task might have been too easy, prompting minimal tool use. Therefore, we carefully defined our recruitment criteria.

Participants were eligible if they met at least one of the following criteria: (1) completion of at least two ASL courses or (2) at least two years of ASL learning experience. All participants signed an Institutional Review Board (IRB)–approved consent form and completed a demographic questionnaire before beginning the study.

Participants included 12 females and 1 male, aged between 18 and 34 years. Of the thirteen, ten reported between two and five years of ASL learning experience. Twelve participants identified as hearing, while one identified as Deaf or Hard of Hearing (DHH). Ten participants had completed ASL II, while others were either currently enrolled in ASL III or had taken ASL I with more than two years of ASL learning experience. Most participants had prior experience using online ASL dictionary tools (e.g., *Handspeak* [38], *Signing Savvy* [39], and *ASL Bloom* [5]) and reported high comfort with using new technology. Their familiarity with AI-based sign language learning systems was reported as "low."

4.2 Experimental Study Guide

Procedure. All sessions were conducted in person in a meeting room within a lab. Participants interacted with the prototypes on a 14-inch MacBook M3. We began by introducing ourselves, explaining the study tasks, and obtaining consent to record the participant's screen. Participants were then introduced to the prototypes (S&S and S) for both conditions, followed by a demonstration of how to use each tool for the assigned tasks through a pre-recorded tutorial video, which did not include the four stimulus videos used in the

Table 2: Recognition performance across datasets and segmentation conditions (mean ± standard deviation).

Dataset	Data Setting	Segmentation	Top-1	Top-2	Top-5	Top-10	Top-20
ASLLRP [50]	Broader Corpus	AI-Segmented	0.33 ± 0.30	0.39 ± 0.34	0.44 ± 0.38	0.48 ± 0.40	0.51 ± 0.42
		Human-Segment	**0.46 ± 0.41**	**0.50 ± 0.44**	**0.52 ± 0.46**	**0.54 ± 0.46**	**0.56 ± 0.46**
	Stimuli Videos	AI-Segmented	0.61 ± 0.15	0.68 ± 0.14	0.78 ± 0.15	0.88 ± 0.11	0.91 ± 0.07
		Human-Segment	**0.90 ± 0.06**	**0.95 ± 0.06**	**0.95 ± 0.06**	**0.97 ± 0.06**	**0.97 ± 0.10**

main user study. We presented both prototypes at the start of the session to establish a consistent understanding of the study workflow. For the *Search-Only* (S) prototype, participants were told to select single-sign spans to ensure they understood the functional boundaries of the tool. However, to evaluate manual versus automatic segmentation realistically, we did not correct participants if they struggled with precise sign boundaries, e.g., due to coarticulation.

We then confirmed participants' comprehension of the procedure before beginning the tasks.

Participants were provided with two video-based search prototypes corresponding to each condition: (1) the *Segment-and-Search* prototype (S&S), which included AI-based automatic segmentation for multiple signs, and (2) the *Search-Only* (S) prototype, which supported only single sign lookup. After piloting and consultation with our team's ASL educator, four example videos were selected from ASLLRP [50] and trimmed to approximately 10 seconds each, containing two to three signed sentences. These videos have been described in detail in subsection 3.2. Each participant viewed two videos per prototype, resulting in four tasks total. The order of conditions as well as the videos was counterbalanced using a balanced Latin square design [17].

During each task, participants watched the assigned video and produced an English translation using the given prototype. They were instructed to either translate the videos into English or transcribe them using ASL glosses. We did not intervene when they produced a mix of the two, as both practices are common in classroom assignments. After completing each video task, participants filled out a questionnaire containing the following items:

NASA TLX. Participants completed the NASA-TLX questionnaire [21, 22] to assess perceived workload across six standard subscales (Mental Demand, Physical Demand, Temporal Demand, Effort, Performance, and Frustration). Each item was rated on a 21-point scale ranging from "Very Low" to "Very High." We computed an aggregate score afterward.

Search Satisfaction Questions. Two questions were included to understand participants' perceived satisfaction with the search experience.

(1) "*I am satisfied with the search experience.*" This 5-point Likert-scale question (ranging from "Strongly Disagree" to "Strongly Agree") was adapted from prior research on search systems [31].
(2) "*I am satisfied with the way the results are ranked.*" This question (ranging from "Strongly Disagree" to "Strongly Agree") was adapted from information-retrieval studies [2] and has been applied in ASL dictionary research [24].

Perceived Usefulness Question. An additional question was included to assess participants' holistic perception of each tool's usefulness.

(1) "*How useful were the search results?*" This 5-point usefulness-scale question (ranging from "Not at all" to "Very") was adapted from prior work evaluating the perceived usefulness of search systems [2].

Participant interactions were logged through Firebase[8], including button activations (e.g., *Play-selection*, *Search*, and *Segment-and-Search*), navigation actions in the search results, and submitted transcriptions. These interactions were then analyzed quantitatively and qualitatively.

After completing all tasks, participants took part in a semi-structured interview focusing on their overall experience and perceptions of the tool's quality with the focus on automatic-segmentation and search functionalities. All interviews were audio-recorded and later transcribed. Only one member of the research team coded the interviews, but the codes and their interpretations were discussed collectively. We primarily used deductive thematic analysis [53] to answer our question regarding the characterization of differences in participants' use of various UI elements in both prototypes.

Entire study sessions were scheduled for 70 minutes but typically lasted about 80 minutes. Each participant received $40 as compensation.

5 Experimental Study Findings

We report findings from both statistical analyses of the Likert data and analyses of observational and qualitative findings to compare outcomes and behaviors across the two conditions.

5.1 NASA TLX and Search Experience

To compare participants' responses to subjective feedback questions, including perceived workload (NASA TLX), search experience, and system usability, we conducted within-subject statistical analyses. Thirteen participants in our experimental study each completed two video translation tasks, yielding $N = 26$ paired observations of participant–video pairs per condition[9]. We used

[8]Firebase: https://firebase.google.com/

[9]We acknowledge that analyzing responses at the participant–video level does not fully account for within-participant dependence because each participant contributed two observations per condition. Consequently, the resulting confidence intervals and significance tests should be interpreted cautiously. The sequence of conditions and video assignments were Latin balanced [65], and all videos were of comparable length, drawn from the same dataset, and judged by an ASL educator to be of similar complexity. These controls reduce potential confounding from condition order and video assignment but do not eliminate within-participant dependence entirely. We therefore treat these analyses as exploratory estimates of differences between conditions rather than definitive participant-level inferences.

paired-sample t-tests to assess mean differences and Wilcoxon signed-rank tests to verify robustness against non-normality. To quantify the precision of these estimates, we report 95% confidence intervals for all mean differences. For all non-significant results, we additionally conducted two one-sided equivalence tests (TOST; [37]) to determine whether observed effects could be interpreted as practically equivalent, using an equivalence margin of ±0.5 standardized mean difference at $\alpha = .05$.

Among the NASA-TLX subscales, ***Mental Demand*** and ***Temporal Demand*** **scores were significantly lower for the segmentation and search condition.** ***Mental Demand*** ($M_{diff} = -2.40$, 95% CI $[-4.35, -0.45]$, $t(25) = -2.52$, $p = .018$, Cohen's $d_z = -0.46$). ***Temporal Demand*** ($M_{diff} = -1.17$, 95% CI $[-2.27, -0.07]$, $t(25) = -2.17$, $p = .039$, Cohen's $d_z = -0.40$)

All other TLX subscales (*Effort*, *Frustration*, *Physical Demand*, and *Performance*) showed trends favoring the *Segment-and-Search condition*. *Performance* scores, which are reverse-coded such that higher values indicate better outcomes, were also higher for this condition. However, none of these differences reached statistical significance.

We further analyzed the aggregated NASA-TLX score and found that **overall workload while translating videos using the *Segment-and-Search* condition was significantly lower than with *Search-Only*** ($M_{diff} = -6.27$, 95% CI $[-11.59, -0.95]$, $t(25) = -2.41$, $p = .023$, $d_z = -0.44$). Figure 3 visualizes the 95% confidence intervals for all subscales and the aggregate.

Figure 4 visualizes the 95% confidence intervals for all Search Experience and System Usability items. Participants reported **significantly higher satisfaction with the search experience in the *Segment-and-Search* condition compared to *Search-Only*** ($M_{diff} = 0.50$, 95% CI $[0.06, 0.94]$, $t(25) = 2.35$, $p = .026$, $d_z = 0.43$).

On average, participants reported slightly higher satisfaction with and the usefulness of the search results ($M_{diff} = 0.03$, 95% CI $[-0.22, 0.28]$) in the *Segment-and-Search* condition compared to *Search-Only*.

5.2 Translation Quality

We compared performance between conditions using fine-grained and holistic metrics. Eight participants produced sentence-level translations, while five used glosses or a combination of both. To account for differences in video duration and sign count, we normalized all sign-level measures by the number of signs in each source video. Descriptively, participants using the *Segment-and-Search* prototype produced a higher proportion of accurate signs (66.8% vs. 58.9%) and fewer omissions (33.2% vs. 41.1%). The proportion of additional signs was comparable between conditions (45.1% in the *Segment-and-Search* condition vs. 44.2% in the *Search-Only* condition). More details about the analysis can be found in Appendix B.

We also computed a holistic translation-quality score (out of 10) for each video in collaboration with an ASL educator, following the procedure described in [23]. The educator assigned each score without considering the experimental condition. A paired-samples t-test found no significant difference between conditions ($p = .46$), although scores were numerically higher in the *Segment-and-Search* condition ($M = 7.04$, $SD = 2.33$) than in the *Search-Only* condition ($M = 5.49$, $SD = 2.26$).

Across the next three subsections, we present findings from deductive thematic analyses of open-ended data collected through post-interviews, observational notes taken during the interviews, and automatically logged data from Firebase.

5.3 Impressions of Benefits and Performance of Both Prototypes

Like prior work [10], participants emphasized that web searches are often unhelpful for ASL lookup. P5 noted, "*with ASL, you can't really... just like Google a word.*" They viewed the video comprehension tool as filling an important gap by enabling sign search and recognition directly from videos. P1 contrasted it to their usage of feature-based dictionaries: "*it was like a different kind of dictionary... I basically described... the five parameters of... the sign... there's no dictionary [like this] really, for... ASL. So it's definitely... helps [to be in the] context.*"

Participants described learning situations where such a tool would be valuable. P4 shared, "*for homework... [course instructor] would sign something and take a video of herself... I think if I had this tool, it would have been useful.*" Similarly, P6 said it could help "unblock" comprehension, noting, "*a lot of the time when I was learning, I'd get videos... and I would just be completely lost... this definitely allows an opportunity to be like, hold up... let's make sure I know this sign.*"

However, some participants reflected on potential downsides. P1 observed, "*I think it would probably lose part of... the culture of the language. But I think as a learning tool, it's really important.*" P4 similarly cautioned that "*I would have worked less hard to learn the language... it would tell me the answers. Like ChatGPT.*"

While answering various questions related to both prototypes, participants rated the system's performance (of finding the sign in the list) between 60% and 80% accuracy. P1 estimated, "*Let's say like 80%, 75%. There are some things that still didn't help, but it helped a big percentage of forming and understanding signs.*" Similarly, P2 reported, "*I was able to find probably 80% of the signs I was looking for... a few I couldn't find, but most of them were there, even if not always at the top of the list.*" P3 added, "*I feel like it had about 60–70% accuracy, but you had to make sure the clips were cut very short and concise to get the specific movements.*"

Some participants noticed mismatches between confidence values and correctness. P5 noted, "*It would be like 90% confidence, but then it would have nothing to do with the sign... and sometimes it was 0% but was actually correct.*" Despite this, P6 found that the relevant results were usually presented early: "*When it had the result, it had it up front... it was nice and easy.*"

Reflecting on the role of human expertise, P9 noted that "*Someone who's possibly more knowledgeable than I could identify those more easily... whereas the AI does it a little bit more efficiently,*" suggesting that while the AI supports efficient retrieval, more experienced users may better recognize subtle similarities between signs.

Participants also perceived the segmentation performance positively. P3 estimated that "*95% of them were cut really well. Some of them were a little off, but it was easier to visualize.*" However, a few users observed that pre-segmented clips sometimes missed full

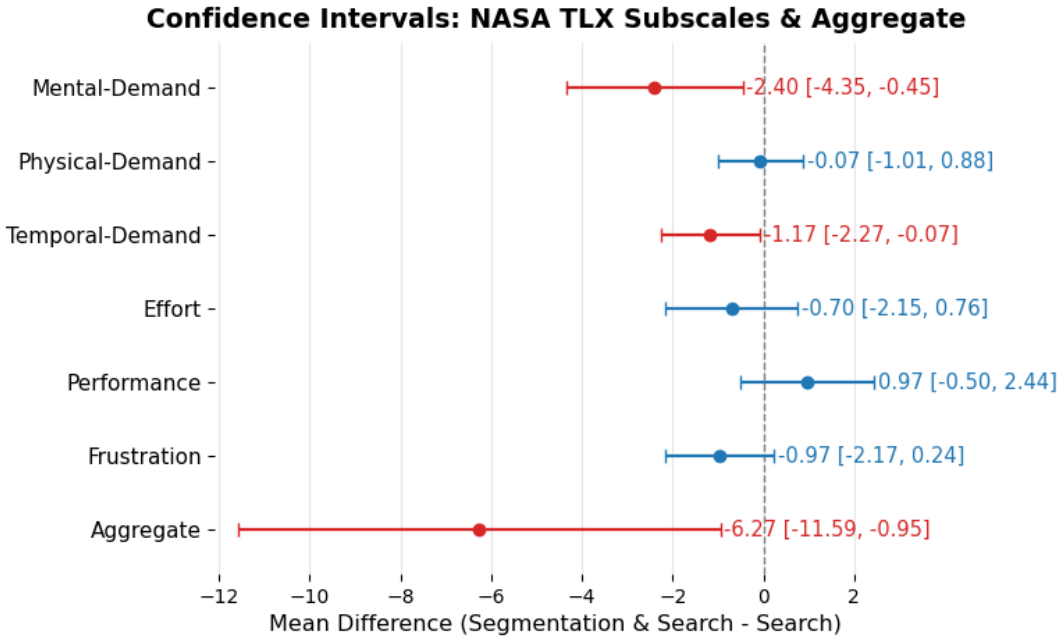


Figure 3: Paired differences with 95% confidence intervals for NASA TLX subscales and the aggregate workload score. Negative values indicate lower perceived workload under *Segment-and-Search* compared to *Search-Only*. For `Performance` sub-scale positive value indicates higher perceived performance under *Segment-and-Search* condition compared to *Search-Only*. Red line denote statistically significant differences ($p < .05$).

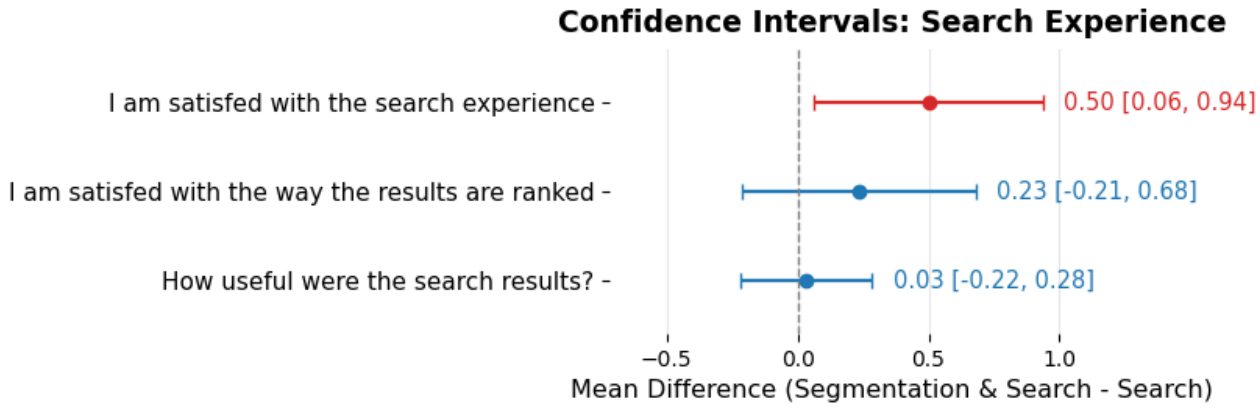


Figure 4: Paired differences with 95% confidence intervals for the Search Experience and Usefulness and System Usability metrics. Positive values indicate higher ratings for *Segment-and-Search* compared to *Search-Only*. Red line denote statistically significant differences ($p < .05$).

signs: "*Sometimes the segments would cut the sign in half, so it wasn't the full sign.*" (P5)

5.4 Comparing the Use of Segmentation & Search and Search-Only Prototypes

5.4.1 Preference and Usability. Participants generally preferred the *Segment-and-Search* condition version, appreciating aspects such as *convenience*, *time savings*, and *reduced manual effort*.

P2 observed, "*I think I liked the first version (S & S) more because when I had a larger segment, it would show me the difference... and then it would show each one [results in segments] versus the second one.*" Others emphasized that pre-segmentation saved time and simplified navigation. P3 shared, "*The first version was easier because it was pre-segmented for you, so you could easily look at the clips.*" P5 elaborated, "*When it pre-segmented..., it saved me a lot of time... it was definitely easier, not having to segment it myself. Especially*

for someone [with less experience], knowing that individual signs are split up is very convenient."

Participants also described the Segment-and-Search prototype as more engaging and easy to use as well. P6 remarked, "*The AI version was a lot nicer and it helped make it easier to just click on each single word... it made it more enticing to use.*" P11 also added that "*[I prefer the] AI one (S & S)... It just made it quicker to find words. I also felt like it was more accurate. And finding the specific signs.* "

Some users primarily used the segmentation feature for just constraining the spans further, even when they knew the signs: "*It's just easy to navigate; shortening the video wasn't difficult. I feel like that's what I used it mostly for.*" (P2)

Participants reported that manual segmentation, in contrast, required more precision and effort. P2 explained, "*You had to kind of measure it pretty precisely to get the exact sign you were looking for.*" P5 was quick to identify that shortening the span selection will lead to better results through trial and error: "*I realized there was a learning curve... the smaller the segment, the better it was, but if I made it too small, then it wouldn't pick up anything.*"

However, at least four participants did express favorable opinions and pointed out that the *Search-Only* prototype allows them more flexibility and control. P4 noted, "*I enjoyed the customizability of the first tool (S) because I could choose which signs I wanted to see... it was intuitive and not too complicated.*" P10 also described that "*[I prefer] the second one (S), when you do it manually, it was a bit easier because you narrowed down to where you saw the sign.*" P13 mentioned "*So, ... overall consistency, I feel like the other one (S) was a little better.*"

5.4.2 Use of Span-Selector. We analyzed the duration, sequence, and qualitative patterns of spans across the two interface conditions.

Commonalities. Beyond lookup, participants across both conditions also used the span selector for review and comprehension. As P2 explained, "*Besides looking for the sign, being able to use the segment sections to just watch the short part of the video and rewatch that to figure out the sign before even searching*" helped reinforce their understanding and confidence.

Participants described their experience as engaging and creative. Several noted that the activity resembled media editing work—"*it did feel like editing, like videos.*" (P1). In both conditions, participants frequently used the SPAN SELECTOR PLAY HEAD to constrain the timeline and work sequentially on smaller video segments, often after first viewing the entire video. Figure 5 illustrates this pattern.

While some participants adopted a holistic viewing strategy, other participants followed a more incremental approach. Participants progressively trim and segment the video from the beginning, carefully review each portion and conduct searches step by step. Participant P10 exemplified this pattern (see Figure 6).

Some participants also revisited earlier sections that had caused confusion after completing their initial viewing, often to verify their translations. As P4 noted, "*I always like to go back and double-check.*" Others appreciated the tool's ability to distinguish between similar-looking signs: "*You could look and see if there were similar ones... when two signs were the same but meant different things... it just made translating a video easier.*" (see Figure 7).

Differences. Table 3 presents the span lengths for different actions and conditions. Overall, most actions exhibited comparable median span lengths between the *Segment-and-Search* and *Search-Only* conditions, particularly for update, play, and search actions. The large standard deviation indicates considerable variability in how participants replayed or reviewed video segments, suggesting differing strategies for regaining context or verifying signs. In contrast, participants using the *Segment-and-Search* condition worked with shorter, more consistent spans.

We also observed that participants in the *Search-Only* condition often inferred unknown signs from surrounding context, requiring them to rewatch longer segments or even entire videos before returning to a focused selection. As P7 noted, "*I was just looking for the sign that made sense with the sentence.*"

Finally, the timing of search activity varied notably between conditions. In the *Segmentation and Search* condition, searches typically occurred mid-translation, often after identifying a segment, suggesting deliberate integration of search into the translation process. By contrast, in the *Search-Only* condition, searches occurred either early, sometimes before playback, or late in the task for verification (See Figure 8).

5.5 Design Suggestions

Participants proposed a few improvements to enhance usability and learning value of the comprehension tool.

P1 suggested incorporating **more targeted filtering as a post-search step**, noting, "*A narrower filtering system... maybe to incorporate how, like, the five parameters.*"

A few participants also recommended adding **feedback features that generate signing based on translation written**. P3 remarked, "*I wish there was like a text-to-sign aspect that I could check my work with... but I know that's not really the purpose of the software.*"

P5 proposed **integrating a video-based search** option to support users who already know a sign, explaining, "*If you know the sign, adding a feature where you could just, like, search the sign also instead of having to pin it... sometimes I knew what the sign was, but I wanted to make sure. So if I could, like, Google that sign in the platform, that would also be helpful.*"

Others emphasized the need for **adjustable playback speed** to improve comprehension of fast signing. P3 said, "*I would have the ability to slow down the clip so I could watch it slower*". P8 mentioned "*I wish you could slow down the video,*" and P6 similarly appreciated having "*control to watch in slow motion, step by step, which is kind of nice.*"

Participants also recommended including a **replay button** to quickly rewatch clips without waiting for them to finish. P3 noted, "*I wish there was a button to return it back to the beginning, even if it hasn't reached the end.*" Others suggested increasing the variety of sign models, with P2 stating, "*Maybe get more variety in the people that are your models for the signs,*" and reducing lag during playback and search, as P1 mentioned, "*It lagged sometimes, so that could be annoying.*"

Some participants also recommended adding **preset segment durations** or drag controls to reduce repetitive actions when working with preferred video span lengths. For example, P3 said, "*Maybe*

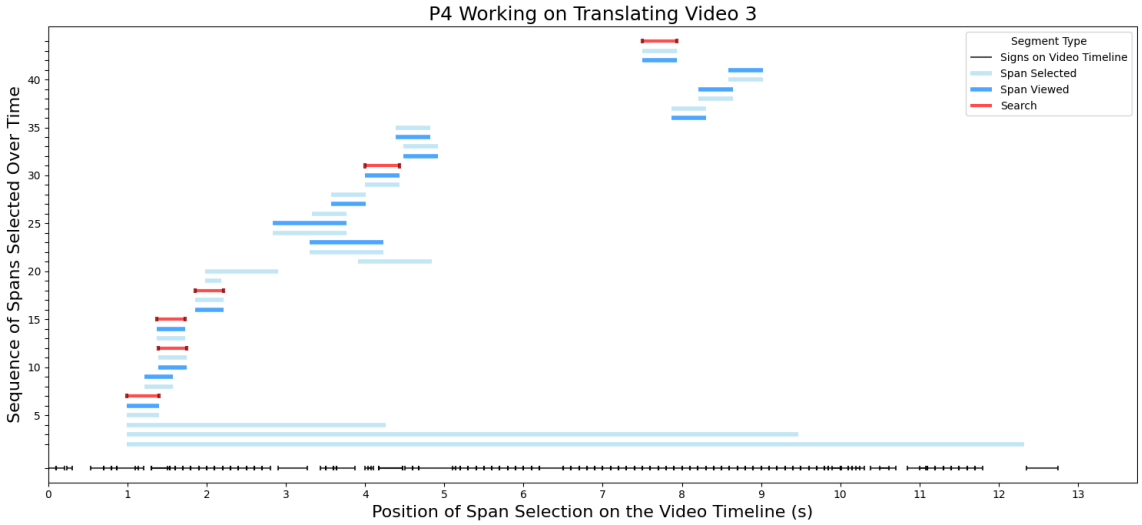


Figure 5: Participant P4 starts with the entire video and then subsequently views narrower segments, using the Search function to do sign look-ups as they proceed. The black line at the bottom shows placement of actual signs.

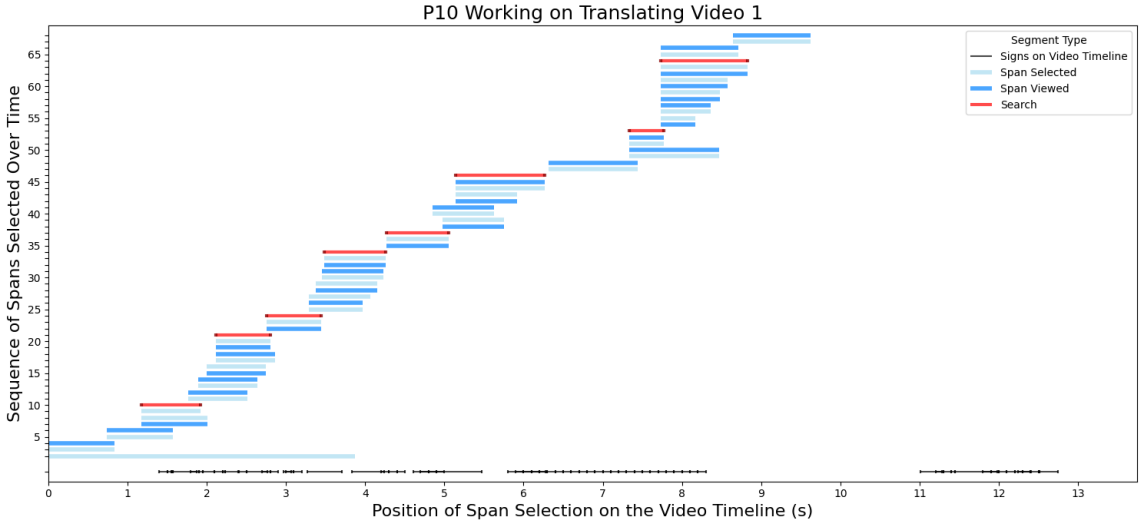


Figure 6: Participant P10's interaction on Video 1 shows a sequential left to right progression. The participant incrementally trims, selects, and plays short spans while performing search actions in a step-by-step manner. The interaction is characterized by fine-grained span selections and frequent searches at each stage, suggesting a strategy of localizing and verifying individual signs before moving forward along the timeline.

start off with, like, one second, like a preset, so it's not so much clicking and dragging."

A few participants expressed the need for a **fallback search mechanism** when target signs are not available in the AI suggestion lists. In such cases, they preferred having an option to externally search and verify the correct sign. P11 noted, "*I think it would be [good] if there could be also like a search option*". Also, P12 mentioned, "*...adding, like a search feature where you can type in a word and then it can generate results so you can kind of compare...*"

Overall, participants liked the presentation of confidence but suggested providing a **color-coded visualization of confidence levels** for better interpretability. P2 proposed, "*Maybe you could do, like, a color code—red, yellow, green—instead of the number.*"

6 Discussion

There has been a substantial amount of research in both HCI-based sign language learning technologies, e.g., dictionary systems [10, 23, 26, 36] and games [6, 61], and AI methods that enable such

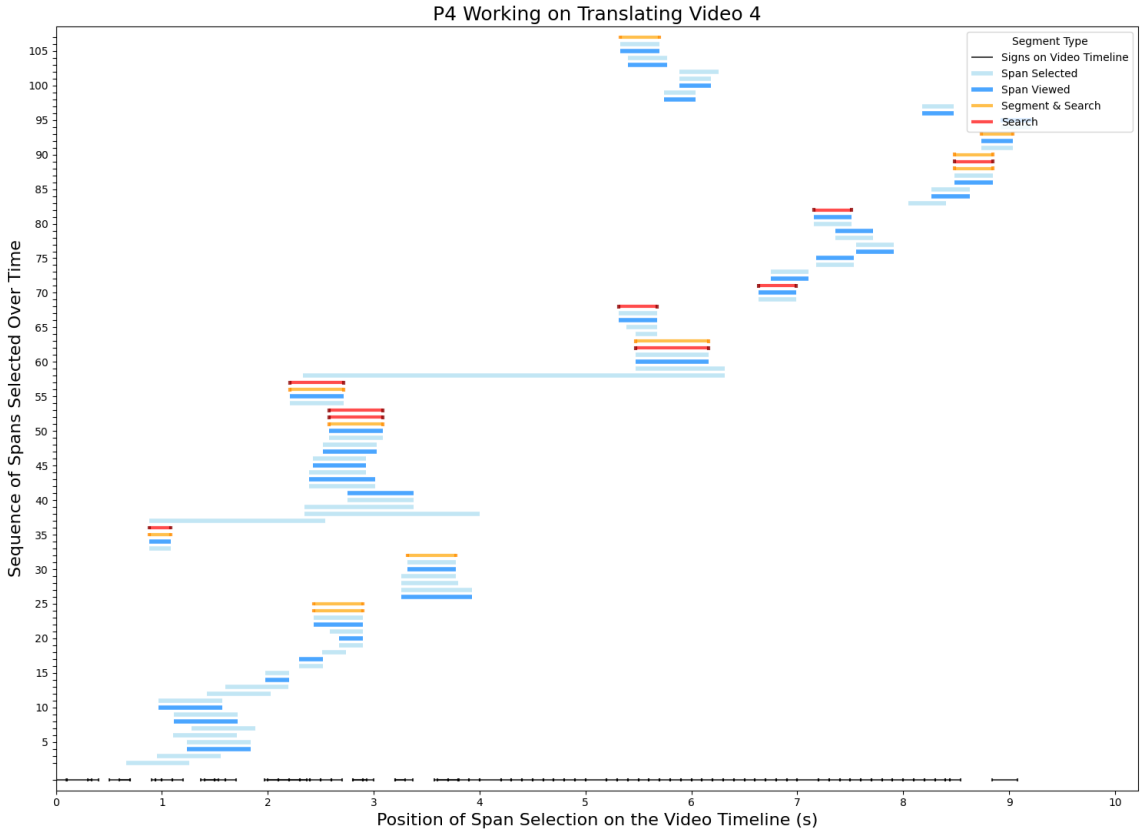


Figure 7: P4 repeatedly alternates between watching, segmenting, and searching across the video timeline. The frequent red and yellow indicators suggest active use of both *Search-Only* and *Segment-and-Search* to refine understanding. Towards the end, P4 goes back to working on an earlier segment of the video.

Table 3: Median span length (median ± standard deviation) across actions in both conditions.

Action	Segment-and-Search (Median ± SD)	Search-Only (Median ± SD)
Update Selection	0.84 ± 3.75	0.74 ± 4.17
Play Selection	0.34 ± 2.61	0.59 ± 2.49
Segmentation and Search	3.14 ± 5.27	–
Search	0.47 ± 1.26	0.46 ± 0.40

applications [14, 75]. However, limitations in each domain have constrained the development of fully automatic sign language learning systems [9]. As a result, much of the existing research has focused on smaller-scale systems consisting of applications of isolated sign recognition for small vocabulary sets. Recent advances in automatic isolated sign recognition [14, 75], supported by larger datasets and Transformer-based architectures, represent a significant shift in technical capability. More recently, emerging work on automatic sign segmentation [47, 56, 62] introduces new possibilities for integrating these models into learning contexts, although this potential remains largely unexplored. Our work builds on these recent developments and represents an initial effort toward designing more comprehensive, large-vocabulary learning systems that support a key aspect of sign language learning: video comprehension.

We build on recent work exploring in-situ sign lookup in video-based learning. We implement two AI-enabled prototypes: *Segment-and-Search*, which applies recognition to automatically generated segments, and *Search-Only*, which applies recognition to user-selected spans. Technical performance was lower than in the original benchmarks [14, 47, 75], although these comparisons involve different datasets and evaluation conditions. Segmentation errors reduced downstream recognition, particularly at top-1, while the difference narrowed at top-20. Because the four study videos were selected for reasonable model performance and included in final

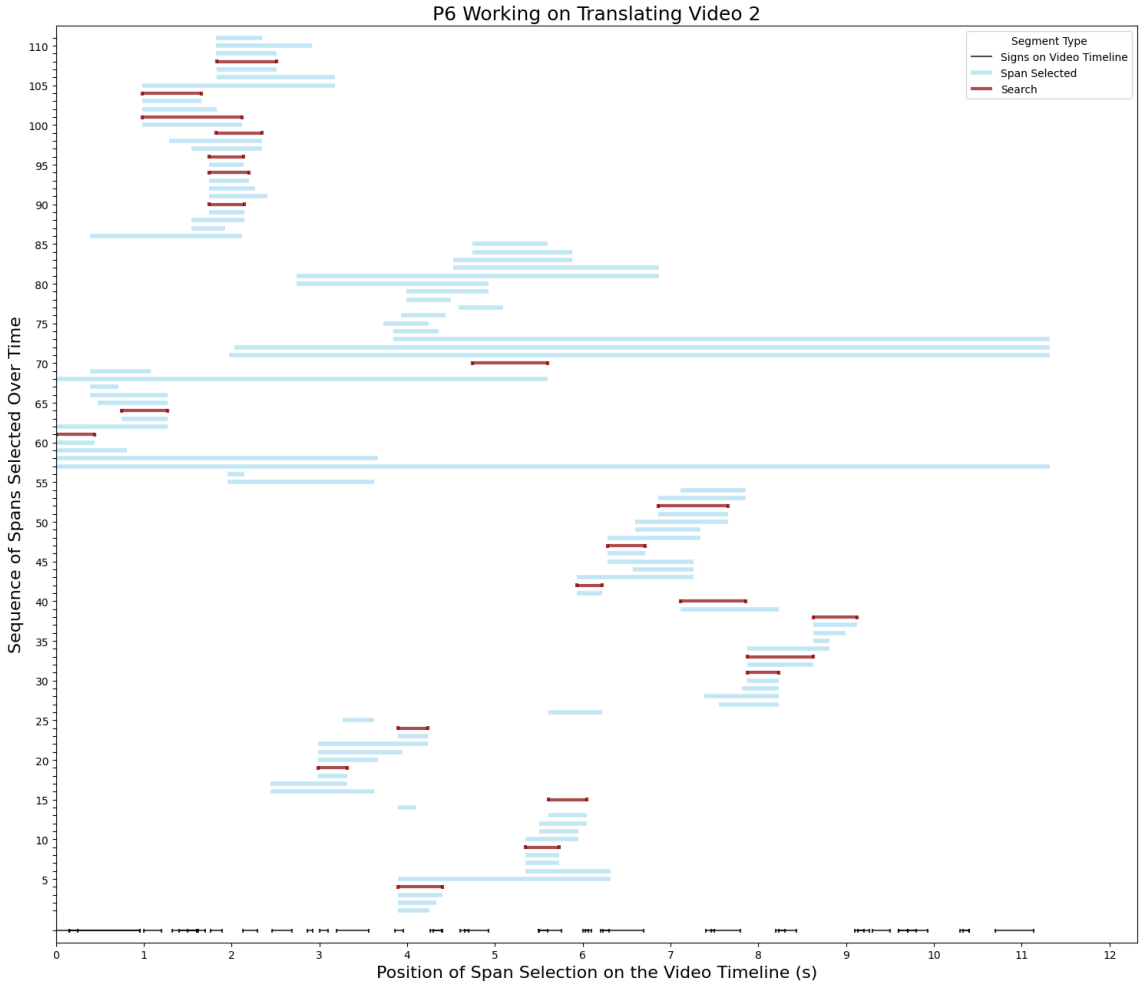


Figure 8: Participant P6's interaction on Video 2 shows a non-linear pattern. The participant repeatedly alternates between short, localized selections and longer spans that cover larger portions of the video timeline. Frequent search actions indicate a strategy of verifying and/or refining translations across multiple passes. Sometimes the participant also expands around a capture, perhaps to capture contextual information. There are larger gaps between subsequent named signs at times as shown in the black line at the bottom. The gaps consist of pauses or other non-named linguistic constructions.

fine-tuning, our findings reflect interaction under relatively favorable conditions and should not be interpreted as evidence of general model performance on more complex ASL videos.

We then compared two versions of the prototype across multiple metrics of performance and usability to answer our research questions.

To address RQ1, which examines perceived workload, we analyzed NASA-TLX ratings across conditions. Participants reported significantly lower overall workload when using the *Segment-and-Search* interface, including a significant reduction in *mental demand*. Together, these findings suggest that segmentation lowers the perceived effort involved in video comprehension tasks, even as automatic segmentation introduces new interface interaction dynamics.

To address RQ2, which examines satisfaction, we found that participant satisfaction was significantly higher with the *Segment-and-Search* system.

To address RQ3, which focuses on translation quality, we found a marginal improvement in holistic translation quality with the *Segment-and-Search* interface. Prior work comparing user-selected segment-based search with *search-by-feature* dictionary interfaces reported higher translation quality with segment-based search [23, 27]. Our findings extend this line of evidence by suggesting a potential benefit of automatic segmentation within a *search-by-video* system.

The answer to RQ4, concerning participants' use of the prototypes, is more nuanced. Both interfaces influenced how participants

approached video viewing and translation. With the *Segment-and-Search* interface, participants reported greater confidence and comfort in analyzing sequential portions of videos, often working with more uniform segment lengths even before activating segmentation or search. In contrast, participants in the *Search-Only* condition showed greater variability in segment lengths, frequently selecting the entire video or revisiting longer spans to recover context or missing signs when search results were insufficient. Consistent with prior work on this condition in comparison to *search-by-feature* baseline [23], participants also spent more time refining shorter spans before searching. In our follow-up interviews, some participants described this slower, more deliberate process as cognitively demanding but beneficial for deep engagement and attention to sign order.

6.1 Recommendations for Researchers & Designers

Our analysis of qualitative data also revealed new design recommendations for ASL video comprehension tools, many of which may extend to video-based dictionary applications as well.

- DG Consistent with prior work on ASL dictionaries [24], participants emphasized the value of incorporating linguistic feature-based post-filtering to refine imperfect search results and help users constrain outputs based on specific sign parameters.
- DG Participants expressed a desire for finer control over playback speed. While prior work has identified challenges related to fast signing [23], our findings further support the inclusion of adjustable video speed. In addition to playback adjustments, participants recommended features such as a replay button to quickly rewatch specific segments and average drag or preset span controls to minimize repetitive actions when selecting similar segment lengths.
- DG Our findings highlight the value of receiving feedback on the quality and scope of their selected segments. Some learners were uncertain whether a chosen span was too short to capture a complete sign or too long to maintain precision. Future systems could provide visual feedback on span length, e.g., using color cues that indicate when a selected segment exceeds an optimal duration for sign recognition. In addition, confidence visualization could be extended beyond recognition results to reflect the reliability of both segmentation and search outputs, helping learners make more informed decisions, e.g., when to refine their selections and when to trust system output.
- DG Participants suggested the integration of an in-situ, video-based ASL dictionary as a "fallback mechanism" for scenarios where a sign in continuous video appears similar to a known citation form but differs slightly, e.g., due to coarticulation. Such functionality would allow users to cross-check and confirm uncertain signs without leaving the video context. This can also help learners practice both citation-form signs and their continuous signing variants that have undergone coarticulation.
- DG Participants envisioned future AI systems capable of generating signing feedback based on their transcriptions. In practice, this could involve avatar-based visualizations that render users' written translations into signing, enabling comparison with original videos. Prior HCI research on expressive learning tools has demonstrated interest in feedback that reuses a learner's recorded submissions, e.g., using face-swap technology [29] or using pose-aware LLM-based approaches [35]. Our findings reveal a related but distinct opportunity: *supporting comparison by generating signing output from learners' translations.*
- DG Our findings suggest the importance of aligning interface complexity with the learner's stage and goal. The *Segment-and-Search* interface was advantageous on workload and satisfaction measures, lowering workload and increasing users' confidence in analyzing larger, more consistent spans of signing. In contrast, the *Search-Only* interface encouraged more deliberate segmentation, closer scrutiny of fine-grained movements, and broader contextual exploration when identifying individual signs. Such behavior may benefit learners at earlier stages of ASL acquisition who are still developing sensitivity to linguistic sub-units. These results suggest that, rather than one interface universally outperforming the other, the optimal configuration may depend on whether the learning objective prioritizes efficient comprehension or analytic practice. The possibility of distinct PRACTICE and ASSIST modes represents a preliminary design direction raised by one participant and echoed indirectly in related feedback from others. Further research is needed to evaluate whether and how these modes should be implemented, as well as to understand how learners and signers naturally segment continuous signing.

6.2 Insights for Research in Related Human-movement Contexts

Beyond sign language video comprehension, span-based segmentation and search can also support video annotation tasks involving human movement data. In fact, emerging research on sign language understanding has begun to explore such approaches, enabling annotators to retrieve glosses, linguistic features, or related examples directly from video segments rather than sifting through glossaries to identify correct glosses [49].

Building on this trend, our findings suggest broader opportunities for intelligent interfaces that foster interactive engagement with videos containing named semantic components or segments that represent meaningful units of action, language, or behavior. The span-based segmentation and lookup paradigm demonstrated here could extend naturally to other structured human-activity domains, such as dance choreography [15, 28], sports training [74], instructional videos in healthcare, e.g., surgery [71], or even other leisurely activities, e.g., bread-making tutorials, where specific gestures or steps correspond to procedural or other types of meaningful units.

Video span selector as a UI element has already been adopted in various contexts, for instance, in video or audio editing tools and on platforms like YouTube [72], which allow users to trim and reshare

short clips from longer videos. In light of our findings, such applications could integrate automatic segmentation with direct lookup or comparison to afford powerful learning or annotation opportunities to achieve similar outcomes, e.g., more reflective observation, reduced cognitive load, etc. In this sense, our work contributes to a growing design space of segment-aware video interaction tools and illustrates how tightly coupled in-situ segmentation and retrieval can transform passive viewing into an active, inquiry-driven learning experience.

7 Limitations and Future Work

There are several limitations of our work that can serve as motivation for future work.

ASL is a linguistically complex natural language with its own grammatical organization [63]. It exhibits regional variation and includes complex constructions, such as classifier constructions [3] and compound signs [69]. Consequently, ASL cannot be fully represented as a sequence of discrete, context-independent lexical signs. Our findings should therefore be interpreted only as evidence about learners' interaction with automatic segmentation and sign lookup in this specific setting. They should not be interpreted as evidence that the system captures the broader linguistic complexity of continuous ASL signing.

The sign segmentation model we employed scales to multiple sign languages [47] but was only largely evaluated on the Public DGS Corpus of German Sign Language (e.g., [20, 52]). We conducted an additional evaluation on ASL, specifically using our own video set, and observed that performance decreased slightly but remained stable overall. As new segmentation models emerge and are trained and evaluated more extensively on ASL recognition tasks, future researchers will be able to examine how these advancements further improve the use of automatic sign segmentation to improve ASL learners' video comprehension.

Relatedly, our AI pipeline was developed to support a prototype system for investigating the use and utility of automatic segmentation, rather than to benchmark the underlying models. We used state-of-the-art isolated-sign recognition datasets [14, 50] and models [14, 75], fine-tuning the recognition model on segmented data. We anticipate that the pipeline's performance would degrade substantially on more complex videos, as evidenced by our initial analysis of the broader corpus. Future work could improve recognition performance through fine-tuning on larger and more diverse datasets or by leveraging emerging foundation models, such as *SignX* [16] and *SignGemma* [18].

Recruiting ASL learners for this study posed notable challenges, which necessitated certain methodological choices. These included adopting statistical approaches suited for smaller sample sizes when analyzing Likert-scale data and relying more heavily on observational evidence to identify behavioral patterns. Most ASL learners possess only beginner-level skills, often acquired through online resources or introductory courses, and are not yet capable of watching continuous signing and producing written transcriptions. Conversely, advanced learners, such as those in interpreting programs, may find the task too simple. While the tool could support learners across proficiency levels, we focused on intermediate learners, who are most likely to benefit from in-situ sign lookup during video comprehension. Future research should expand participation to include a wider range of ASL learners to better capture the diversity of learning needs and experiences.

Relatedly, our study was conducted in a lab-based setting to control for potential confounding variables. Future research could pursue more ecologically valid studies, where tools like this are deployed over time with learners or integrated into classroom curricula.

Our videos were selected in consultation with an ASL educator, but represent only a small portion of the types of sign language videos that ASL learners encounter during their studies. Similar to prior WoZ work [27], future research should adapt these systems to support a wider array of complex and specialized video types. This includes multi-person interactions (e.g., two-person dialogues), which introduce overlapping signs and camera occlusion, as well as artistic or non-standard formats like song signing, which features more classifiers, heightened prosody, fluidity, and non-lexical movements. Expanding to significantly larger vocabulary and diverse signing speeds will also be important for real-world generalization.

8 Conclusion

Learning from video is a central component of sign language education, yet few tools effectively support ASL learners in video-based learning. Recent advancements in AI for sign language understanding, including automatic sign recognition and segmentation, offer new opportunities to enrich this experience. In this work, we present and release a prototype video-comprehension tool that integrates segmentation and sign lookup capabilities, enabling learners to identify unfamiliar signs while watching and translating ASL videos. Our findings from an experimental study with 13 ASL learners highlight the benefits and usage patterns of AI-assisted features. We also propose design guidelines for future sign language learning technologies and discuss broader implications for human–AI interaction in the context of human movement recognition systems.

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A Procedure of measuring translation quality and analysis

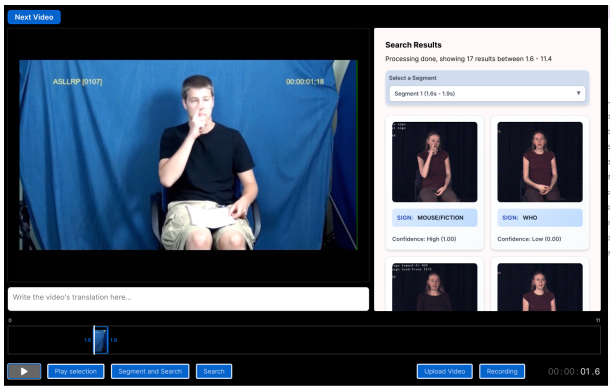
We compared performance between conditions using fine-grained lexical-content metrics. We derived the reference sign tokens from the original stimulus annotations and our educator team members transcripts. Classifier constructions, pointing and possessive markers, and the fingerspelled name were excluded from the lexical-sign denominator, resulting in 8, 16, 13, and 8 reference unique glosses across the four videos.

We accepted inflectional variants, common spelling errors, and conservative semantic equivalents as matches (e.g., *saw*, *noticed*, or *found* for SEE). Each response word could match only one reference token, and repeated reference tokens required separate evidence in the response. Articles and other English grammatical scaffolding in translated responses were not counted as additional signs. All measures were normalized by the number of included lexical sign tokens in the corresponding source video. The table below summarizes the results.

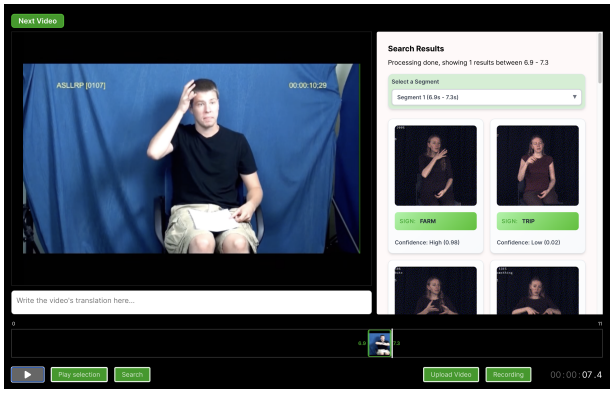
Table 4: Descriptive analysis of student-written translations and transcriptions

Metric	Segment-and-Search	Search-Only
Accurate signs	66.8%	58.9%
Missing signs	33.2%	41.1%
Additional concepts	45.1%	44.2%

B Segment-and-Search prototype and Search prototype



(a) Segment-and-Search prototype (S&S)



(b) Search-Only prototype (S)

Figure 9: Segment-and-Search (a) and Search-Only (b) prototypes. For Search Only prototype the navigation bar only shows one sign.